Tecnológico Nacional de México
Campus Cd. Madero.

Ingeniería en Electrónica.

Programación Estructurada.

Evaluación del curso.

- Individual
- Proyectos y prácticas 100%
    - Se suben a Github (Repositorios privados y con invitación)
    - El trabajo se evalúa en base a cumplimiento de requerimientos
    - Posteriormente hay una evaluación personal y puede modificar la nota del trabajo

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Prácticas:

1. Radar Speed Traps

A highway-patrol speed-detection radar emits a beam of microwaves at a frequency designated as f_e . The beam is reflected off an approaching car, and the radar unit picks up and analyzes the reflected beam's frequency, f_r. The reflected beam's frequency is shifted slightly from f_e to f_r because of the car's motion. The relationship between the speed of the car, v, in miles per hour (mph), and the two microwave frequencies is:

$$v = (6.685 \times 10^8)\left(\frac{f_r - f_e}{f_r + f_e}\right) mph$$

where the emitted waves have a frequency of f_e = 2 × 1010 sec^{-1}. Using the given formula, you'll write a C++ program, using the software development procedure, to calculate and display the speed corresponding to a given received frequency f_r.

2. Temperature conversion

a. Write a C++ program that accepts a number followed by one space and then a letter. If the letter following the number is f, the program is to treat the number entered as a temperature in degrees Fahrenheit, convert the number to the equivalent degrees Celsius, and display a suitable message. If the letter following the number is c, the program is to treat the number entered as a temperature in degrees Celsius, convert the number to the equivalent degrees Fahrenheit, and display a suitable message. If the letter is neither f nor c, the program is to display a message that the data entered is incorrect and terminate. Use an if-else chain in your program and make use of these conversion formulas:
Celsius = (5.0 / 9.0) × (Fahrenheit - 32.0)
Fahrenheit = (9.0 / 5.0) × Celsius + 32.0

3. Count of voltage ranges

An engineer has constructed a two-dimensional array of real numbers with **n/3** rows and **3** columns. This array currently contains test voltages of an amplifier. Write a C++ program that interactively inputs ***n*** (multiple of 3) array values, and then determines the total number of voltages in these ranges: less than 60, greater than or equal to 60 and less than 70, greater than or equal to 70 and less than 80, greater than or equal to 80 and less than 90, and greater than or equal to 90. A function must be included that receives the voltage array, the lower limit, and the upper limit of the interval to be checked.

b. Entering **n** voltages each time in the program written runs is cumbersome. What method could be used for initializing the array during the testing phase?

c. How might the program you wrote for this exercise be modified to include the case of no voltage being present? That is, what voltage could be used to indicate an invalid voltage, and how would your program have to be modified to exclude counting such a voltage?

4. Voltage intervals and statistics

Modify the previous program so that, using functions, we can find:

a. minimum voltage
b. maximum voltage
c. average voltage
d. standard deviation of the voltage readings

5. Power calculation with arrays and functions

Write a program that stores the multiple resistance values in an array named resistance: 16, 27, 39, 56, and 81, by example. Your program should also create two arrays named current and power, each capable of storing five double-precision numbers. Using a for loop and a cin statement, have your program accept five user-input numbers in the current array when the program is run. Your program should store the product of the values of the squares of the current array and the resistance array in the power array. For example, use power[1] =

resistance[1] * pow(current[1],2). Your program should then display the following output (fill in the chart):

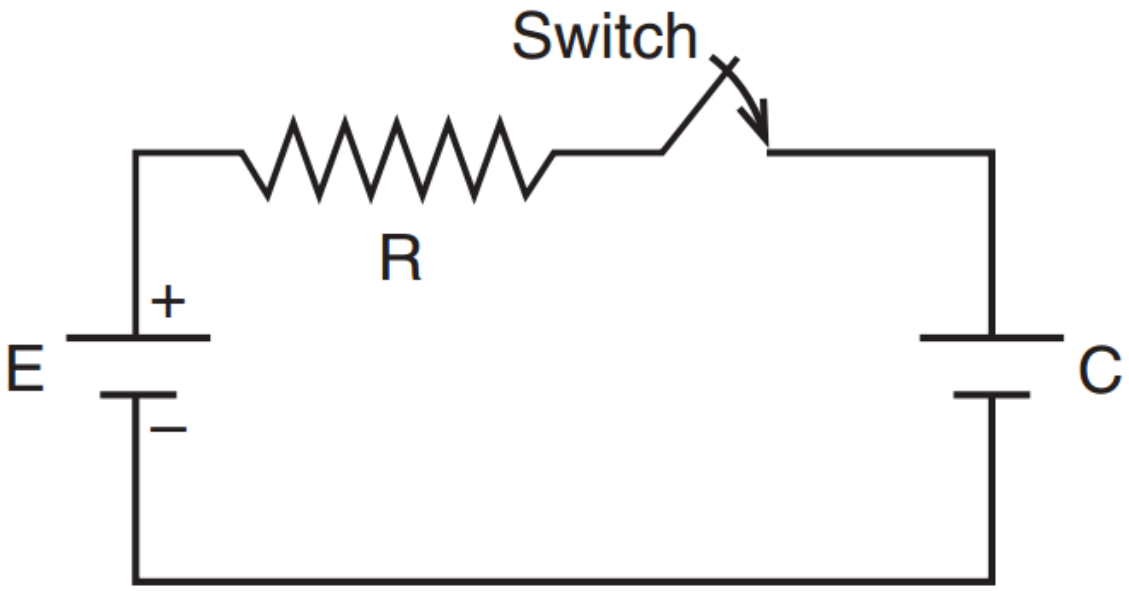
Resistance	Current	Power
16		
27		
39		
56		
81		
Total:		

6. RC circuit

When the switch shown in Figure 7.14 is closed at time t = 0, the voltage, V, across the capacitor, C, is given by this formula:

$V = E(1 - e^{-t/RC})$

E is the voltage of the battery.
R is the resistance in the circuit.
C is the value of the capacitance.
e is Euler's number (2.71828 . . .).
t is the time in seconds.



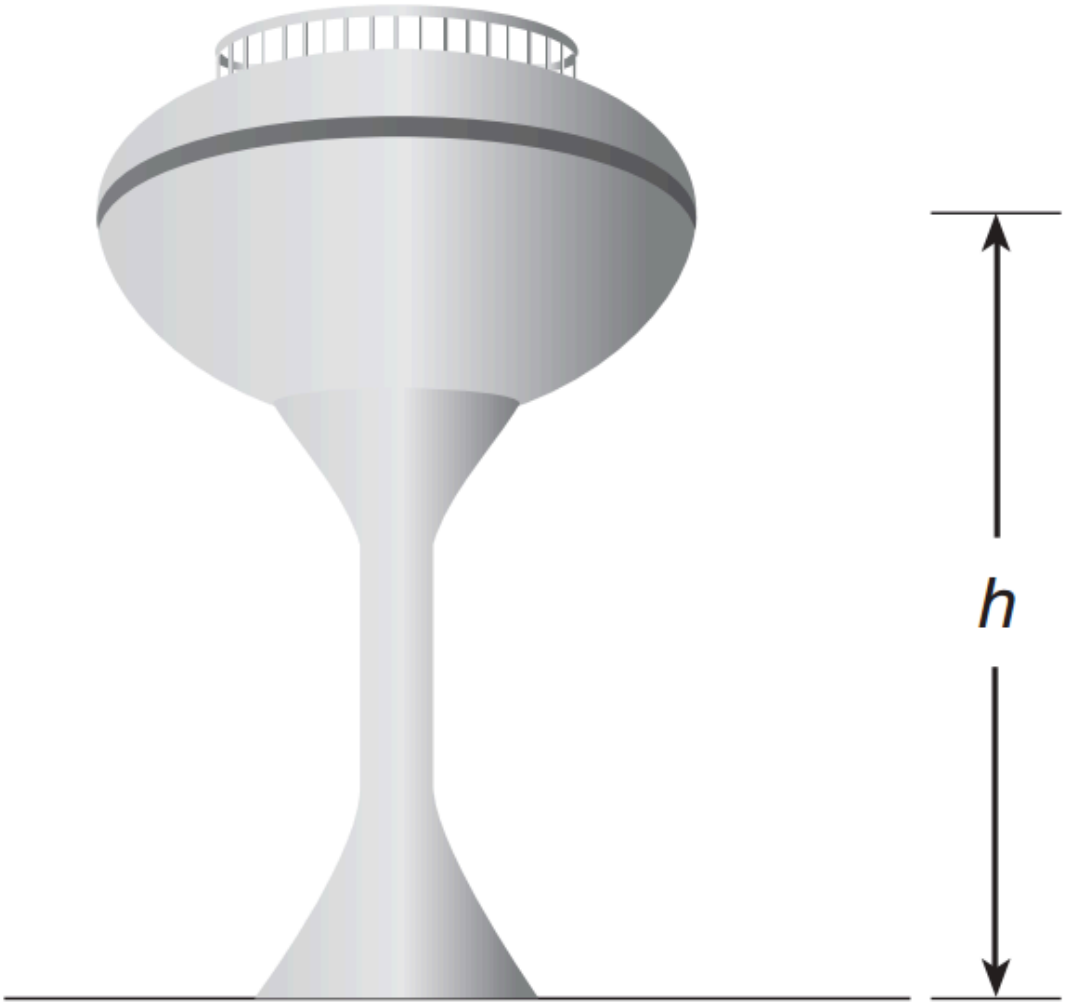
Provide values for E, R, and C, write a program to plot the voltage across the capacitor from t = 0 to t = final value in increments of given seconds.

7. Hydraulics

Water towers are used to store water and provide water pressure to households, industrial plants, and offices. (Typically, this water pressure averages from 60 to 80 lb/in2.) The

pressure in the supply pipe at the base of the water tower (see Figure below) is given by this formula:
$P = (\rho \times g \times h)k$

P is the pressure at the base of the tower (lb/in2).
ρ is the water density (1.94 slug/ft3).
g is the acceleration caused by gravity (32.2 ft/sec2).
h is the water height in the tank (ft).
k is the conversion factor (1 ft2/144 in2).



Using this formula, write a program to plot the water pressure in a pipe at the base of the water tower for water heights of h_i to h_f feet in increments of inc feet. The pressure should be displayed in units of lbs/in2 (psi). Generally, municipal water pressures average from 60 to 80 psi, so the program must inform of trouble designs.

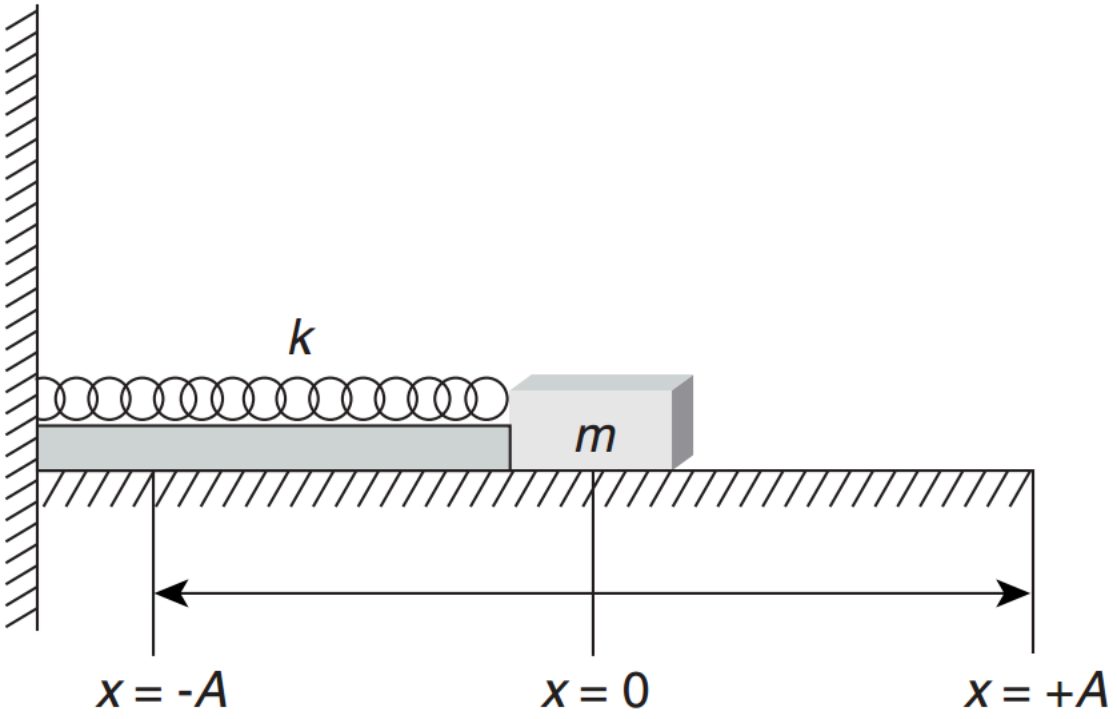
8. Physics

Figure below shows a harmonic oscillator, which consists of an object of mass, m, attached to one end of a spring. The spring's other end is attached to a wall, and the object is free to slide over a frictionless surface. Assuming the object is initially at rest (that is, the spring is

neither stretched nor compressed) and then pulled to position A at time t = 0, the displacement of the mass at any other time, t, is given by this formula:

$x = A \cos (t \sqrt{k/m})$

x is the displacement.
k is the spring constant (N/m).
m is the mass (kg).
A is the initial displacement (cm).



For givens A, k and m, write a program to plot the displacement of the mass from t = i to t = f seconds in increments of inc seconds.